Name_____ Date_____ Period_____

LESSON 2-8 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Finding the Mean

Maryland Standard 6.SP.B.5c · Learning target – I can find the mean of a data set by sharing the total equally among the values.

REMEMBER

- The **mean** is a fair share: add all the values, then divide by **how many** values there are.
- **Mean = sum ÷ number of values.** Count every value, including repeats and zeros.
- The mean is between the least and greatest values. It does not have to be in the data set.

WORKED EXAMPLE

Kiara read 12, 9, 15, 8, and 11 pages on five nights. Find the mean.

Step 1 Add: 12 + 9 + 15 + 8 + 11 = 55

Step 2 Count: there are 5 values.

Step 3 Divide: 55 ÷ 5 = 11. Check: 11 is between 8 and 15.

Answer **11 pages**

WORD BANK

mean – the fair share: add, then divide

sum – the total when you add

data set – all the numbers you collected

PRACTICE

show your work · label units

1. Find the mean of each data set.

a. 6, 9, 12 _____

b. 15, 20, 25, 40 _____

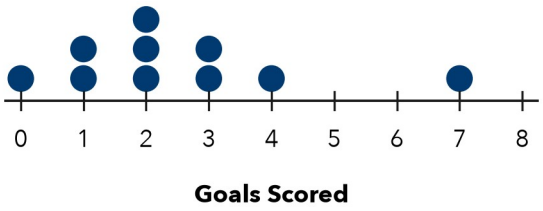
c. 7, 0, 5, 8 _____

d. 3, 3, 9, 4, 6 _____

e. 2.5, 3.5, 4.5 _____

f. 18, 22, 14, 26, 30 _____

2. **Soccer Goals.** The **dot plot** shows the **goals** a team scored in each game. Each dot is one game.



Goals Scored

a. Total goals: _____

b. Number of games: _____

c. Mean goals per game: _____

3. **Rainfall.** Rain fell on **five** days: 0.4 in, 1.2 in, **0 in**, 0.9 in and 0.5 in. Add all five, then divide by **5**.

Show your work.

Answer_____ in

4. **New Value.** The mean of 8, 10, and 12 is 10. A fourth value, 20, is added. What happens to the mean?

   ☐ It goes up. ☐ It goes down. ☐ It stays the same.

   New mean_____

5. **Find the Error.** To find the mean of 8, 0, 6, and 10, Darnell added to get 24. Then he divided by 3 because "zero does not count." Explain his error and find the mean.

   Frame *Darnell is wrong because zero ___. The mean is ___.*

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask 4 people at home how many glasses of water they drink in a day. Find the mean.

Name_____ Date_____ Period_____

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Reaching a Target Mean

Maryland Standard 6.SP.B.5c · Learning target – I can find the value a data set still needs in order to reach a target mean.

REMEMBER

- **Target total = target mean × number of values** (count the missing value too).
- **Missing value = target total − sum so far.** Check: the new total ÷ count should equal the target mean.

WORKED EXAMPLE

Jamal's quiz scores are 78, 85, and 90. What score on quiz 4 gives a mean of 85?

Step 1 Target total: 85 × 4 = 340

Step 2 Sum so far: 78 + 85 + 90 = 253

Step 3 Missing: 340 − 253 = 87. Check: 340 ÷ 4 = 85

Answer **87 points**

WORD BANK

target mean – the mean you want

target total – target mean × number of values

missing value – target total − sum so far

PRACTICE

show your work · label units

1. Complete the table.

Step 1 Target total = target mean × number of values.

Step 2 Missing value = target total − sum so far.

	Values	Target mean	Target total	Sum so far	Missing value
A	3	12		25	
B	4		64	49	

2. **Bowling.** Aaliyah bowled **112, 98, 105 and 121**. What score in **game 5** gives her a **mean of 110**?

Step 1 Find the target total: 110 × 5.

Step 2 Add her four scores.

Step 3 Subtract to find game 5.

Answer_____ points

3. Car Wash. A club raised $45, $60, $52, and $38 on four Saturdays. What must it raise on the fifth Saturday for a mean of $50?

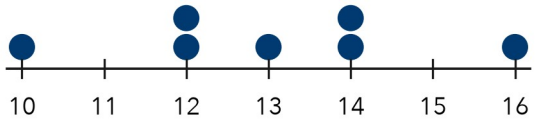
Step 1 Find the target total: $50 × 5.

Step 2 Add the four Saturdays.

Step 3 Subtract.

Answer_____

4. Push-Ups. The dot plot shows how many push-ups Nia did on each of 7 days.



a. Mean for the 7 days: _____

b. Day 8 for a mean of 14: _____

5. Find the Error. Brianna scored 82, 86, and 78. For a mean of 85 on 4 tests, she wrote 85 × 3 − 246 = 9. Explain her error. What score does she need?

Frame *Brianna should multiply by ___ because ___. She needs a score of ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. Four numbers have a mean of 9. Three of them are 7, 10, and 8. What is the fourth number?

A 9 **B** 11 **C** 2 **D** 36

7. A player's points in 5 games have a mean of 14. What is her total number of points?

A 2.8 **B** 19 **C** 70 **D** 9

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___